27. Use rules of inference to show that if $\forall x(P(x) \rightarrow (Q(x) \land S(x)))$ and $\forall x(P(x) \land R(x))$ are true, then $\forall x(R(x) \land S(x))$ is true.

$\boxed{\text{Soln:}}$ $\boxed{\text{Premise:}}$ 1. $\forall x(P(x) \rightarrow (Q(x) \land S(x)))$ 2. $\forall x(P(x) \land R(x))$

$\boxed{\text{Conclusion:}}$ 1. $\forall x(R(x) \land S(x))$

Steps	Reason
1. $\forall x(P(x) \rightarrow (Q(x) \land S(x)))$	Premise
2. $\forall x(P(x) \land R(x))$	Premise
3. $P(a) \rightarrow (Q(a) \land S(a))$	Universal instantiation on 1
4. $P(a) \land R(a)$	Universal instantiation on 2
5. $P(a)$	Simplification on 4
6. $Q(a) \land S(a)$	Modus ponens on 5, 3
7. $S(a) \land Q(a)$	Commutative law on 6
8. $S(a)$	Simplification on 7
9. $R(a) \land P(a)$	Commutative law on 4
10. $R(a)$	Simplification on 9
11. $R(a) \land S(a)$	Conjunction on 10, 8
12. $\forall x(R(x) \land S(x))$	Universal generalization on 11

28. Use rules of inference to show that if $\forall x(P(x) \lor Q(x))$ and $\forall x((\neg P(x) \land Q(x)) \rightarrow R(x))$ are true, then $\forall x(\neg R(x) \rightarrow P(x))$ is also true, where the domains of all quantifiers are the same.

$\boxed{\text{Soln:}}$ $\boxed{\text{Premise:}}$ 1. $\forall x(P(x) \lor Q(x))$ 2. $\forall x((\neg P(x) \land Q(x)) \rightarrow R(x))$

$\boxed{\text{Conclusion:}}$ $\forall x(\neg R(x) \rightarrow P(x))$